A Finite-Center No-Go Boundary for Quotient-Visible Reconstruction

Live review note of June 20, 2026

Abstract

This note states the smallest Team 1 residue in standard language. Let A be the represented von Neumann algebra of supplied observable data, and let $\widehat{A}_q = A \oplus (N \overline{\otimes} \ell^\infty(F_q))$ be a finite central extension with quotient map $\pi_q : \widehat{A}_q \to A$. Any record formed only after applying π_q is constant on the hidden finite central fiber. Therefore no rule whose input is only that quotient-visible record can determine targets that vary on the fiber: hidden central support, ambient faithfulness, finite central cardinality, the predicate $q = 2$, unquotiented tensor coordinates, omitted sector-character fibers, or absolute finite-center entropy calibration.

The proof is elementary: normal quotients of von Neumann algebras are central-projection quotients, and reconstruction from a record is possible only for functions constant on its fibers. The result should therefore be treated as demotion-ready. In statistical language it is a Blackwell/Le Cam/Petz/ Buscemi sufficiency boundary for a declared experiment, channel, subalgebra, or model. The possible contribution is not new operator algebra; it is the black-hole/AQFT warning that recovery, sector, modular, locally covariant, or entropy records reconstruct only what their hypotheses supply. Binary finite rank follows only after a separator, physical-exclusion/maximality premise, finite-alphabet law, or direct measurement channel is added.

Status. This is live non-frozen review metadata. It does not modify either frozen Team 1 paper, Lean archive, public mirror manifest, or SHA-256 package. Its purpose is external classification: known theorem, false statement, missing hypothesis, too broad physically, or useful expository corollary.

1 Question

Standard reconstruction theorems reconstruct supplied structure: a represented algebra, a functor, a sector category, a code algebra, a channel, a split inclusion, a trace-normalized Type II algebra, or a measured spectrum. Team 1's remaining question is the converse:

> If the displayed record is already quotient-visible with respect to a hidden finite central completion, what further premise is required before one may reconstruct hidden finite central rank, ambient faithfulness, unquotiented coordinates, or the binary predicate $q = 2$?

The word "reconstruct" means "is a function of the displayed data on the declared comparison class." There is no obstruction if the model supplies the full finite center, the atomic measurement, or q itself as input. The obstruction is to inferring those targets from records that have already forgotten finite central directions.

2 Same-Data Finite-Center Extensions

Definition 1 (quotient-visible record)**.** Let A and N be von Neumann algebras, let F_q be a finite set with $|F_q| = q$, and set

$$\widehat{A}_q = A \oplus \left(N \overline{\otimes} \ell^\infty(F_q)\right).$$

Let

$$\pi_q : \widehat{A}_q \to A, \qquad \pi_q(a, x) = a$$

be the normal quotient map. A record D_q is quotient-visible if every state, effect, channel, sector label, recovery map, modular datum, tensor coordinate, spectroscopy datum, or entropy quantity used to form D_q factors through π_q. Equivalently, D_q is constant on the kernel fiber $0 \oplus (N \overline{\otimes} \ell^\infty(F_q))$.

Theorem 1 (same-data finite-center obstruction)**.** *Let $\mathcal{C}$ be a comparison class containing at least two quotient maps $\pi_q : \widehat{A}_q \to A$ as above, or hidden refinements of one such finite central summand, with quotient-visible records D_q having the same displayed value after applying π_q. Then no reconstruction rule whose input is only that displayed record can determine any target that is not constant on the comparison fiber. In particular it cannot determine:*

1. *whether the represented algebra A was ambient-faithful;*
2. *the nonzero finite central support in the hidden summand;*
3. *the hidden finite central cardinality q;*
4. *the predicate $q = 2$;*
5. *unquotiented tensor coordinates in the hidden finite central summand;*
6. *omitted central-character sector fibers; or*
7. *an absolute finite-center entropy calibration attached to that summand.*

Those targets become determinate only after adding a datum that separates, excludes, identifies as null or gauge, calibrates, or directly measures the hidden finite central directions.

Proof. The kernel of π_q is the normal central summand

$$0 \oplus (N \overline{\otimes} \ell^\infty(F_q)) \,.$$

More generally, ultraweakly closed two-sided ideals in von Neumann algebras are central-projection summands, so ordinary normal quotients forget precisely a central summand. Since D_q factors through π_q, all hidden choices in that summand give the same displayed record. But the targets listed in the theorem vary over same-record completions: take $q = 2$ and $q = 3$, vary the hidden central support, refine an omitted central-character fiber, change an unrepresented tensor coordinate, or change an additive finite-center entropy normalization while preserving the quotient record. A function of the displayed record must assign the same value to all same-record completions, so it cannot equal a target that differs between them. The only exits are premises that remove the comparison fiber: a separating readout, a maximality or physical-exclusion theorem, a null/gauge identification, a calibration rule, a finite-alphabet law, or a direct measurement channel. □

Corollary 1 (binary rank is not a quotient-visible consequence)**.** *In the comparison class of Theorem 1, the predicate $q = 2$ is reconstructible from quotient-visible records only if the records are injective on the binary-versus-nonbinary fiber, or if an added physical premise collapses that fiber. Entropy, recovery, sector, modular, or locally covariant data that are identical for $q = 2$ and $q = 3$ do not force binary rank.*

3 Positive Exits

The theorem is a no-go only relative to quotient-visible data. It is killed, demoted, or made inapplicable by any of the following positive hypotheses:

Positive input	Effect
Faithful finite-center readout	Separates the finite-center contrast space; then the accounted rank is recoverable.
Physical maximality or source faithfulness	Says the represented algebra is the full physical algebra, not a quotient of a hidden central extension.
Split, type-I, or tensor-independence hypothesis	Supplies spatial tensor coordinates instead of asking generated-algebra data to reconstruct them.

Full sector category or full code algebra	Makes DHR/DR or OAQEC/JLMS reconstruction apply to the intended object, rather than to an omitted quotient.
Trace or area calibration	Fixes the finite-center entropy normalization.
Finite-alphabet law or direct spectroscopy	Measures the relevant finite fiber and can make $q = 2$ a genuine output rather than an inference from quotient-visible entropy.

4 Demotion Boundary

The mathematical core of Theorem 1 is not a Clay- or Nobel-level new theorem. It is the central-projection quotient theorem plus the elementary fact that functions of a record are constant on record fibers. Blackwell comparison of experiments and Le Cam deficiency theory say the same thing in decision-theoretic language [1, 2, 3, 4]. Petz sufficiency and later quantum comparison theorems say the noncommutative version relative to a declared subalgebra, channel, or statistical model [5, 6, 7, 8].

The possible Team 1 value is therefore a precise black-hole/AQFT claim boundary:

> Recovery, sector, modular, locally covariant, split, spectroscopy, or entropy records reconstruct the object supplied by their hypotheses. They do not, by themselves, certify maximality against hidden finite central extensions or derive binary rank. To force q or $q = 2$, the finite-center separator, exclusion, calibration, or direct measurement premise must be named.

5 Reviewer Question

Is this note merely a known sufficiency/central-quotient corollary under another name, or is the black-hole/AQFT translation a useful claim boundary? The desired external response is one of:

- *known*: cite the exact theorem and demote the note;
- *false*: give the counterexample or hidden assumption;
- *missing hypothesis*: name the positive premise that makes the target reconstructible;
- *too broad*: name the physical axiom always assumed in the intended setting;
- *useful*: say the finite-center no-go is worth writing as an expository reconstruction boundary.

References

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